SURETY

Agent Risk Underwriting for Amex Agentic Commerce

Continuous risk scoring · Dynamic exposure caps · Circuit breakers · Claims-grade evidence

TEAM SURETY

CodeStreet 2026 · Governance Layer for Financial Agents

THE GOVERNANCE GAP

A Promise Without Underwriting

14 APR 2026

ACE Developer Kit

Amex opens agent-initiated purchasing to developers.

→

14 APR 2026

Agent Purchase Protection

Industry-first cover for registered-agent purchase errors.

→

MISSING

The actuarial layer

No exposure pricing. No live caps. No claims evidence.

WHY THE PROMISE NEEDS UNDERWRITING

An open-ended promise is an open-ended liability. Without live scoring, every registered agent carries the same unpriced exposure.

- Disputes resolve after the money has moved, not before.
- Nothing on file proves what the Card Member actually asked for.
- 324M global chargebacks forecast by 2028, up 24% on 2025, before agent errors are counted.

ONE AGENT ERROR

₹17,000

overspent against a stated ceiling

Weeks of dispute.
No record of the original instruction.

Sources: American Express Newsroom, ACE Developer Kit and Agent Purchase Protection, 14 Apr 2026 · Mastercard / Datos Insights, 2025 Global Chargebacks Outlook.

THE SOLUTION

One Loop: Score, Cap, Stop, Prove

A FENCE ASKS

"Is this single action allowed?"

AN UNDERWRITER ASKS

"What is this agent's exposure, and who pays if it is wrong?"

01

SCORE

Continuous risk scoring

Every action re-scores the agent: intent deviation, baseline drift, category entropy, velocity.

XGBoost over a live Kafka feature stream

02

CAP

Dynamic exposure caps

The score sets the ceiling: per transaction, per merchant category, per window, recomputed live.

OPA policy + Redis token buckets

03

STOP

Circuit breakers

Revoke one agent, a class of agents, or the whole fleet. In-flight requests decline immediately.

Three-level breaker, millisecond propagation

04

PROVE

Claims-grade evidence

Each breach assembles intent, cart, merchant and decision trail into a filable Claims Package.

Hash-chained audit in PostgreSQL

Underwriting, not just enforcement.

LIVE SCENARIO

An Agent Error, Resolved in Seconds

DETECTED

Caught in-flight

68% over the stated ceiling and outside the stated category, scored before the charge settles.

PROTECTED

Protection triggered

Agent Purchase Protection applied automatically. Card Member liability set to zero.

PROVEN

Claims Package filed

Investigation engine: intent, cart, merchant and decision trail in 2.3 seconds. Agent capped to ₹15,000.

Claims Package = the investigation engine that turns Amex's protection promise into closed-loop, evidence-backed resolution.

Manual: 7-14 evidence sources, several days

SURETY: **1 Claims Package, 2.3 seconds**

Conceptual comparison - illustrative

TRANSPARENCY

Underwriting You Can Interrogate

DECISION RECORD

DECLINED

Agent	**shop-bot-7**
Transaction	**₹42,000**
Risk score before	**0.42 (Silver)**
Risk score after	**0.78 (Bronze)**
Decline threshold	**0.65**
Exposure limit	**₹25,000 → ₹15,000**

Replayable with the exact model and policy version.

surety · decision output · case SUR-2026-04817

```
DECISION: DECLINE
Agent:       shop-bot-7
Transaction: ₹42,000 headphones
Risk Score:  0.78  (Threshold: 0.65, was 0.42)

DRIVERS:
  +0.20  intent deviation   (68% over ₹25,000 limit)
  +0.10  amount anomaly     (7.9σ above baseline)
  +0.06  merchant entropy   (4 categories in 5 txs)

COUNTERFACTUAL:
  At ₹23,000:  score = 0.48 → ALLOWED  (Silver tier)
  At ₹42,000, 1 category:  score = 0.58 → ALLOWED  (Silver, flagged)
  At ₹42,000, 4 categories: score = 0.71 → DECLINED (Bronze tier)
```

Drivers and counterfactuals are emitted with every decision, not reconstructed after the fact.

Every decision ships with its drivers and counterfactuals. *Mathematically fair. No black box.*

RUBRIC COVERAGE

Complete Mapping to the Brief

REQUIREMENT	HOW SURETY DELIVERS IT	SHOWN IN THE DEMO
Granular permission model	Risk-tiered terms over action type, merchant category, velocity and amount ceiling. Operator-configurable.	Tier change applied live to a running agent
Dynamic spend caps, real time	Recomputed from the live risk score, enforced in-path by OPA with Redis token buckets.	Cap tightens the moment the score moves
Revocation and emergency stop	Three levels: single agent, agent class, whole fleet. In-flight requests decline at once.	Fleet breaker pulled on stage, declines visible
Operator dashboard	Underwriter's console: live exposure, policy editor, decision replay, one-click breakers.	Console driven live, not screenshotted
Testing and optimisation	50-agent simulated fleet with planted rogues; precision, recall and p99 latency reported.	Measured results and a live tamper test

Designed for Sub-10ms Decisions

01 **Agent action** — Ingests ACE: intent intelligence, cart context

→ 02 **Kafka** — Event backbone

→ 03 **Feature extractor** — Deviation, drift, entropy, velocity

→ 04 **XGBoost scorer** — Continuous agent risk score

→ 05 **OPA + Redis** — Allow, cap or decline in-path

→ 06 **PostgreSQL** — Registry and hash-chained audit

→ 07 **React console** — FastAPI + WebSocket live view

Target: <10 ms p99 policy decisions @ 10K agents · designed for sub-10ms p99 enforcement

Stateless enforcement path

The policy decision needs no session state, so it scales horizontally and is designed to hold the latency target under load.

Tamper-evident by construction

Each audit record hashes the previous one. Editing a single row provably breaks the chain.

Aligned to NIST AI RMF

Mapped to the Govern and Manage functions, so the control story survives a real risk review.

Kafka · XGBoost · Open Policy Agent · FastAPI · Redis · PostgreSQL · React

No exotic dependencies: every component is buildable inside the Round 2 window.

References: Open Policy Agent documentation · NIST AI Risk Management Framework (Govern, Manage).

Trust as a Product, Not a Cost Centre

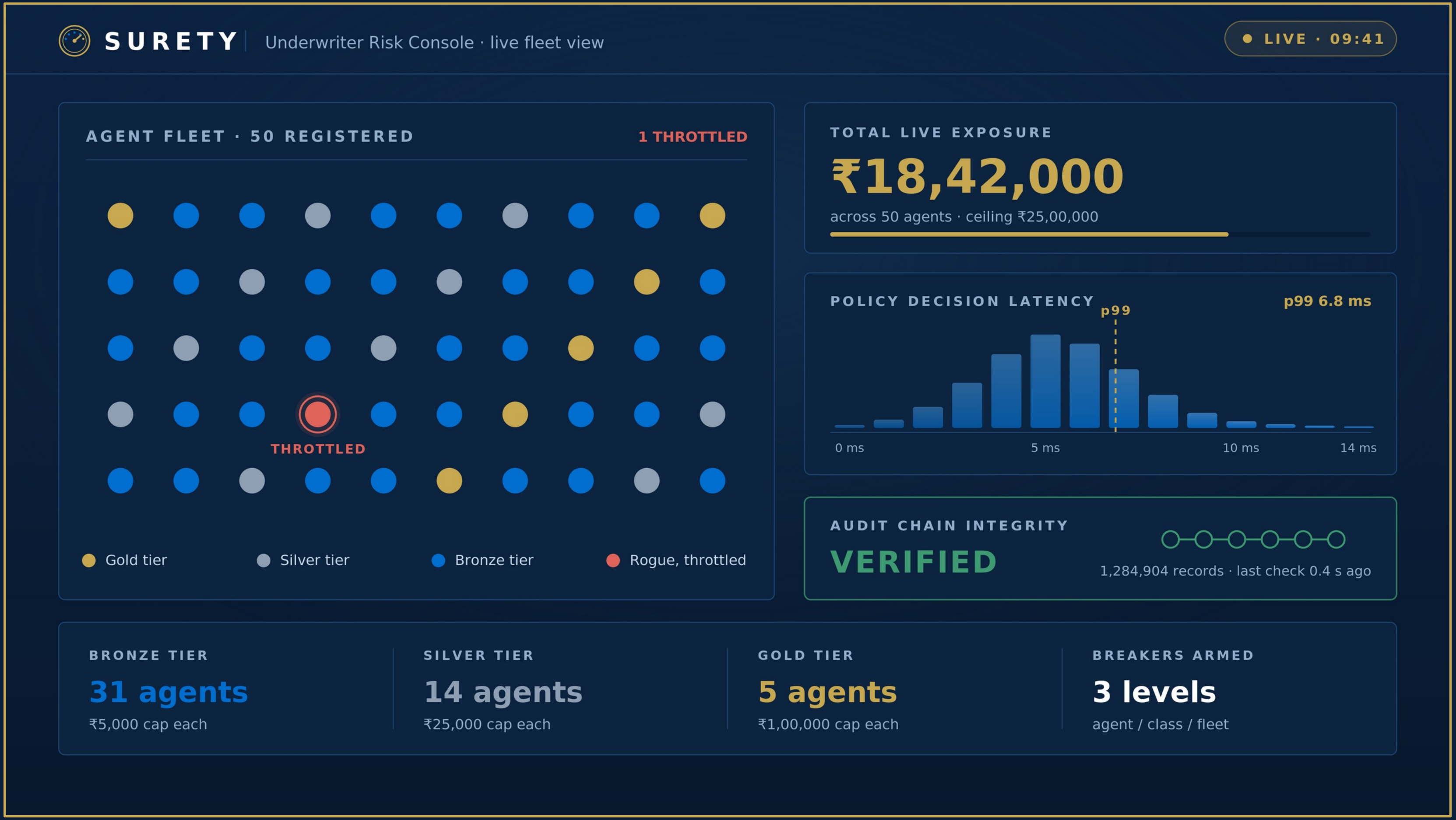
01	**Makes Agent Purchase Protection sustainable**	Every registered agent carries a priced, capped maximum loss instead of open-ended exposure.	Loss ratio becomes forecastable
02	**Cuts the cost of every agent dispute**	Claims Packages arrive pre-assembled and aligned to Amex chargeback reason codes, so resolution starts with evidence in hand.	Lower handling cost per case
03	**Underwriting-as-a-service**	Developers who hold Gold-tier scores for 90 days earn a "SURETY Certified" badge: higher default limits, reduced checkout friction, and a recurring revenue stream for Amex.	Compliance becomes competitive advantage
04	**Extends the Amex trust franchise**	The brand that made card payments safe becomes the brand that makes autonomous payments safe.	Brand-consistent differentiation

Governance stops being a cost of doing agentic commerce. It becomes the product Amex sells.

Why only Amex? Closed-loop network → richer intent signals → better underwriting → faster claims → more trusted agents → more spend → more Amex revenue.

TESTING AND MEASUREMENT

Measured, Not Claimed



ROGUE DETECTION

Precision and recall

50-agent fleet with planted rogue behaviour.

POLICY LATENCY

Target: p99 under 10 ms

Load test against the enforcement path.

BREAKER PROPAGATION

In-flight declines, no leakage

Fleet-wide revoke during live traffic.

AUDIT INTEGRITY

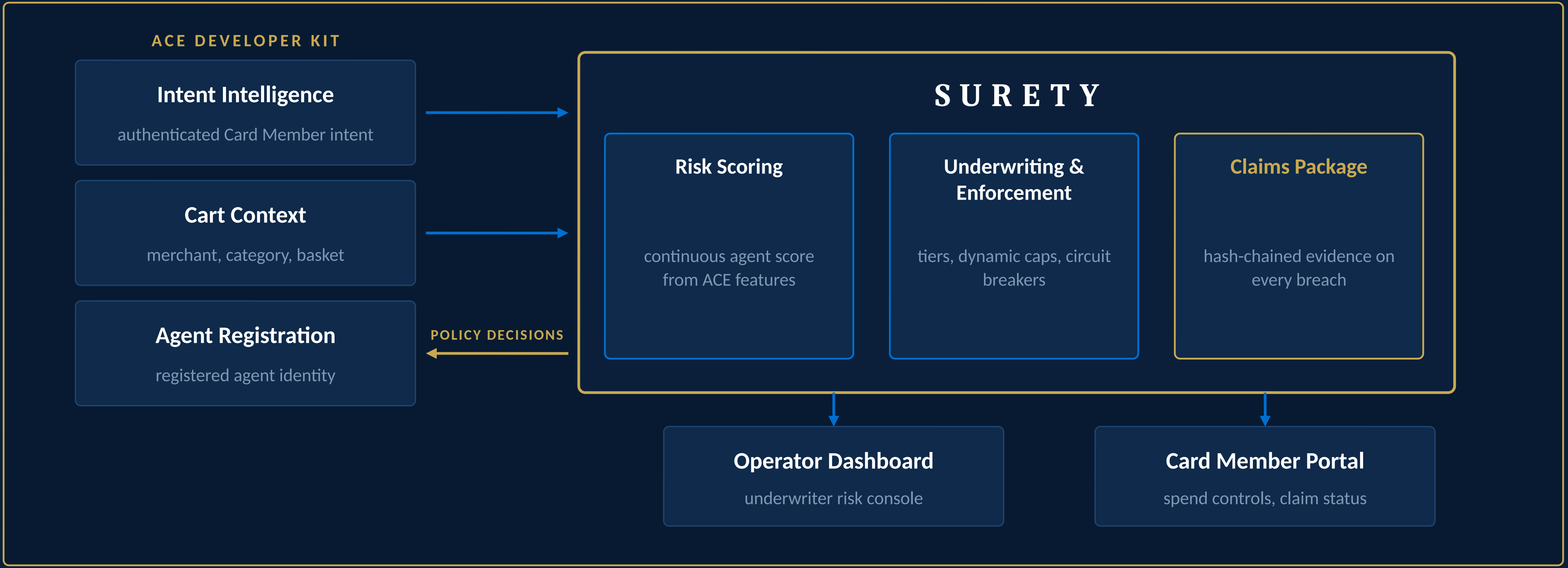
Chain break on screen

A record edited on stage during the demo.

We will not claim performance. We will measure it in front of you.

APPENDIX A

SURETY-ACE Integration Architecture



How SURETY plugs into Amex's ACE kit: Intent Intelligence and Cart Context are ingested as underwriting features; policy decisions are written back to Agent Registration - a closed governance loop.

Every other governance proposal builds a fence.

SURETY builds the actuarial layer.

It decides how high each fence must be, prices the residual risk, and files the claim when it is breached.

The first network to underwrite autonomous transactions at scale owns the moat.

SURETY is the credit score for autonomous agents: a universal trust metric for the agentic economy.

Team SURETY · Thank you